# Overview: MNLI, every split by premise

## One premise, one hypothesis per label

Premise: *You outwitted me.*

- **Entailment (E)** — *I was outwitted.*
- **Neutral (N)** — *It was the first time I've been outwitted.*
- **Contradiction (C)** — *You have never outwitted me.*

## Splits: no premise on two sides

**Tuning** 10% — **CV pool** 90%, 5 folds, repeated 3 times

| | fold 1 | fold 2 | fold 3 | fold 4 | fold 5 |
|---|---|---|---|---|---|

39,359 pairs: settings, early stopping — 353,343 pairs: all comparisons

**Matched** 9,815: same genres (fiction, government, slate, telephone, travel)

**Mismatched** 9,832: new genres (9/11, face-to-face, letters, OUP, verbatim)

Official validation sets: final report only

## Train corpus

| | |
|---|---|
| Pairs | 392,702 |
| Premises (`promptID`) | 128,107 |
| Mean pairs per premise | 3.07 |
| Pairs per label | 33.3% each |
| Mean words, premise | 19.8 |
| Mean words, hypothesis | 10.0 |

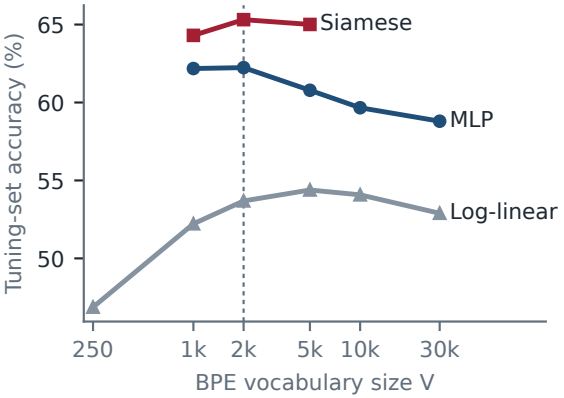
## Choices

| | |
|---|---|
| Tokeniser | BPE, $V = 2000$ |
| Features | Indicator, BoW, BoW + overlap |
| Models | Log-linear, MLP, Siamese MLP |
| Metrics | Accuracy, NLL |
| Compare | Bayesian correlated $t$, ROPE $\pm 1$ pp |

Williams, Nangia & Bowman (NAACL 2018). Test labels are not public.
Code: github.com/miguelalcocker/nlp1-student-group01/tree/main/project1. AI assistance: `project1/AI_USE.md`.

# Q2 Tokeniser: BPE, $V = 2000$, trained on our own pool

## Vocabulary size (one fit per point)




## Training data: is MNLI large enough?

| | |
|---|---|
| Unique premises + hypotheses | 115,349 + 353,343 |
| Running words / word types | 5.8M / 179k |
| `[UNK]` on validation sets | $\leq 0.001\%$ |
| External corpus | none |
| Tuning, validation sets | excluded |

## Other settings (log-linear, BoW)

| Change from case-sensitive BPE | $\Delta$ accuracy |
|---|---|
| Refit inside each fold (CV) | $-0.03$ pp |
| Lowercase, $V = 1$k / 10k | $-0.04$ / $-0.07$ pp |
| Word-level, $V = 1$k / 10k | $+0.04$ / $+0.26$ pp |
| Train / tuning acc., $V = 2$k | 54.7 / 53.7% |
| Train / tuning acc., $V = 30$k | 63.4 / 52.9% |

## What BPE produces

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| `_You` | `_have` | `_never` | `_out` | `w` | `itt` | `ed` | `_me.` |

No spread to draw: settings were not repeated. Tuning-set SE $\approx 0.25$ pp, about the size of a marker. Differences on the tuning set unless marked CV.

# Q3 Features: keep the sentences apart and encode what they share

## The three representations (one input per pair)

| Name | Input | Dims |
|---|---|---|
| Indicator | $[\mathrm{ind}(p);\ \mathrm{ind}(h)]$ | 4,000 |
| BoW | $[\mathrm{cnt}(p);\ \mathrm{cnt}(h)]$ | 4,000 |
| BoW + overlap | $[\mathrm{cnt}(p);\ \mathrm{cnt}(h);\ \mathrm{ind}(p) \odot \mathrm{ind}(h)]$ | 6,000 |

ind: present, 0/1. cnt: count. Overlap: AND of indicators, shaded below.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| *p* | _You | _out | w | itt | ed | _me. | |
| *h* | _You | _have | _never | _out | w | itt | ed | _me. |

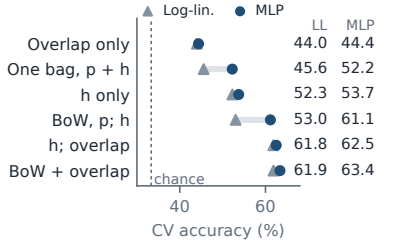
## Weights for contradiction (log-linear, mean of 5 folds)

| Token | Hypothesis block | Overlap block | Hyp. + overlap |
|---|---|---|---|
| _never | +2.10 | −2.47 | −0.37 |
| _no | +1.77 | −1.77 | 0.00 |
| _not | +0.89 | −1.13 | −0.24 |

Shared tokens per pair (mean count): E 10.3, N 7.4, C 7.6; SD 5–7

**E** *You've never had a job.* → *You've never been employed.*

## What each input gives (mean of 15 fits)



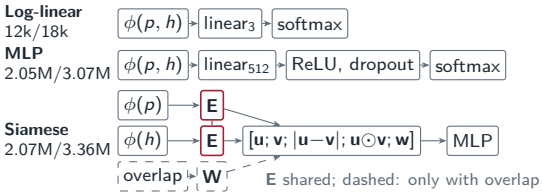

## Alternatives (tuning accuracy, %)

| | Log-lin. | MLP |
|---|---|---|
| BoW | 53.7 | 62.0 |
| TF-IDF per sentence | 53.9 | 63.3 |
| Hashed cross-unigrams | 51.7 | 59.0 |
| BoW + overlap | **62.5** | **64.4** |

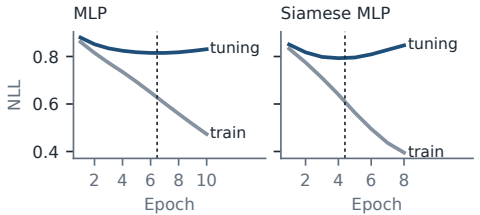
Weight SD over folds $\leq 0.03$; the sum leaves out the premise block. Plot SD $\leq 0.28$ pp; LL: log-linear. Alternatives: one fit each. Cross-unigrams: Bowman et al. (EMNLP 2015).

# Q4 Architectures: from additive scores to an explicit comparison

## Three models (parameters: BoW / BoW + overlap)




## Train and tuning loss (overlap, mean of 15 fits)




## Search on the tuning set

| Setting | Chosen | Alternative: Δ pp |
|---|---|---|
| Learning rate, AdamW | $3\times10^{-4}$ | $10^{-4}$: −0.1 |
| Weight decay, AdamW | 0.3 | 0.1: −0.1 |
| Hidden units | 512 | 128: −0.6 |
| Dropout | 0.3 | 0.5: +0.1 |
| Activation | ReLU | GELU: −0.6 |
| Hidden layers | 1 | 2: −1.2 |
| Log-linear $C$ | 1 | flat from 0.1 |

Mean Δ over 6 neural systems, one fit each (layers: 3 MLPs). tanh: −3.7 pp. SE ≈ 0.25 pp: dropout 0.5 is within noise, −0.1 on the best system.

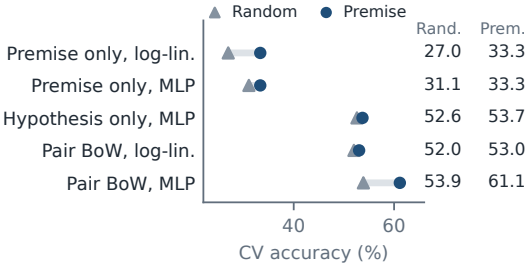
## First guess against chosen setting

| Tuning, % | lr $10^{-3}$, wd $10^{-4}$ | lr $3\times10^{-4}$, wd 0.3 |
|---|---|---|
| MLP, BoW | 60.5 | **62.1** |
| Siamese, BoW | 65.0 | **65.2** |
| Siamese, overlap | 65.5 | **66.3** |

Conneau et al. (EMNLP 2017). AdamW, batch 256, ≤20 epochs, patience 3; log-linear: L-BFGS. Dashed: mean selected epoch. Train loss on 20k pairs, no dropout.

# Q5 Evaluation: the tutorial's random split would mislead us here

## Random against premise split (5-fold mean, seed 0)




## Held-out accuracy, % (Siamese + overlap)

| | Matched | Mismatched |
|---|---|---|
| Single model (mean of 15) | 64.4 | 65.0 |
| Majority vote of 15 | **65.8** | **66.3** |
| Hypothesis-only MLP, vote | 54.0 | 53.3 |
| CBOW (published, test) | 64.8 | 64.5 |
| ESIM (published, test) | 72.3 | 72.1 |
| Hypothesis only (published, test) | 53.9 | 52.3 |

## Where it fails (vote)

| | Matched | Mismatched |
|---|---|---|
| Recall E / N / C | 69.4 / 64.7 / 63.1 | 71.5 / 63.6 / 63.5 |
| Lowest genre | slate 61.2 | verbatim 64.4 |
| Highest genre | government 69.0 | letters 70.2 |

**Hypothesis only** (CV, 53.7 against 64.9): 65% of the best gain over chance

## Other checks

| | |
|---|---|
| MLP − log-linear, BoW: random / premise | 1.9 / 8.1 pp |
| Max \|macro-F1 − accuracy\| in the grid | 0.26 pp |
| Early stopping on fold, not tuning set | −0.01 pp |

Williams et al. (NAACL 2018); Gururangan et al. (NAACL 2018). Binomial SE: held-out 0.5, per label 0.8, per genre 1.1 pp. SD: 15 single models $\leq$ 0.45, folds in the plot $\leq$ 0.7 pp. Selection frozen first.

# Q6 Decision: Siamese MLP; the overlap block is not yet decided

## CV accuracy, % (mean of 15 fits)

| | Log-lin. | MLP | Siam. |
|---|---|---|---|
| Indicator | 53.1 | 61.3 | 64.0 |
| BoW | 53.0 | 61.1 | 63.9 |
| BoW + overlap | 61.9 | 63.4 | **64.9** |

## CV NLL (mean of 15 fits)

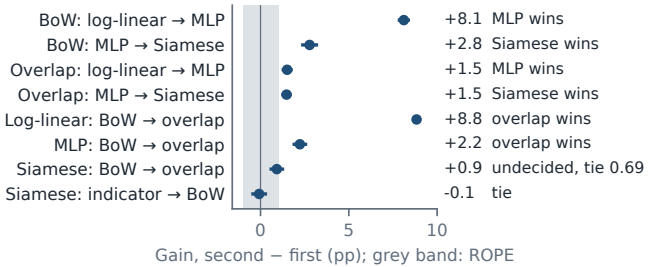
| | Log-lin. | MLP | Siam. |
|---|---|---|---|
| Indicator | .972 | .860 | .816 |
| BoW | .973 | .864 | .820 |
| BoW + overlap | .862 | .826 | **.800** |

SD over fits: 0.16–0.32 pp; NLL .002–.003

## Why ROPE ±1 pp

| | |
|---|---|
| Held-out SE (9.8k pairs) | 0.5 pp |
| SD between CV repetitions | $\leq$ 0.12 pp |
| Benavoli et al. | 1% |

## Paired gain in CV accuracy: posterior mean and 95% interval




## All 36 pairs at ±1 pp

27 decided, 7 ties, 2 undecided. At ±0.5 pp 6 change, at ±2 pp 8 change, none reverses.

Compare with the posterior, not the SDs: fits share data. 15 differences, paired by repetition and fold; each repetition has new folds and a new initialisation. Benavoli et al. (JMLR 2017): correlated $t$, $\rho = 1/5$, `runs=3`. Decided: posterior probability above 0.95.